

EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

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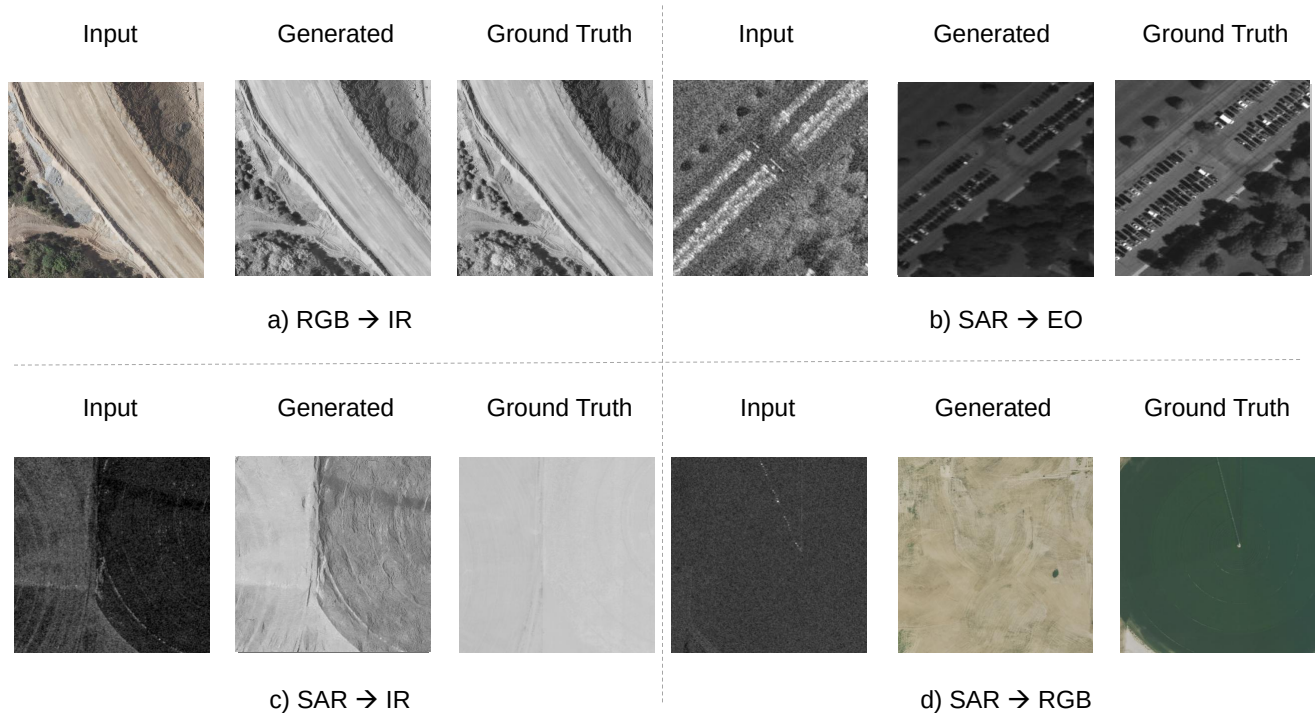


Figure 1. Qualitative results on our hold-out validation set from the official train set. (a) RGB $\rightarrow$ IR, (b) SAR $\rightarrow$ EO, (c) SAR $\rightarrow$ IR, (d) SAR $\rightarrow$ RGB. Each panel shows input, generated output, and ground-truth target. EarthBridge preserves structural integrity while synthesizing high-fidelity textures across Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) modalities.

Abstract

001 Cross-modal image-to-image translation among Electro-
 002 Optical (EO), Infrared (IR), and Synthetic Aperture Radar
 003 (SAR) sensors is essential for comprehensive multi-modal
 004 aerial-view analysis. However, translating between these
 005 modalities is notoriously difficult due to their distinct elec-
 006 tromagnetic signatures and geometric characteristics. This
 007 paper presents **EarthBridge**, a high-fidelity translation
 008 framework developed for the 4th Multi-modal Aerial View
 009 Image Challenge – Translation (MAVIC-T). We explore two
 010 distinct methodologies: **Diffusion Bridge Implicit Mod-**

els (DBIM), which we generalize using non-Markovian
 bridge processes for high-quality deterministic sampling,
 and **Contrastive Unpaired Translation (CUT)**, which uti-
 lizes contrastive learning for structural consistency. Our
 EarthBridge framework employs a channel-concatenated
 UNet denoiser trained with Karras-weighted bridge scal-
 ings and a specialized “booting noise” initialization to
 handle the inherent ambiguity in cross-modal mappings.
 We evaluate these methods across all four challenge tasks
 (SAR $\rightarrow$ EO, SAR $\rightarrow$ RGB, SAR $\rightarrow$ IR, RGB $\rightarrow$ IR), achieving
 superior spatial detail and spectral accuracy. Our solution
 achieved a composite score of 0.38, securing the second po-

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023 *sition on the MAVIC-T leaderboard.*

024 1. Introduction

025 The Multi-modal Aerial View Image Challenge for cross-
026 domain image translation [1, 21, 22] is a leading chal-
027 lenge held in conjunction with the Perception Beyond the
028 Visible Spectrum (PBVS) workshop from 2023 to 2025.
029 Data collected across multiple sensing modalities—such as
030 Electro-Optical (EO), Infrared (IR), and Synthetic Aper-
031 ture Radar (SAR)—provide complementary perspectives on
032 the same scene. However, data coverage limitations and
033 modality-specific constraints often hinder the full utiliza-
034 tion of these diverse sources. For instance, while SAR
035 sensors offer all-weather capabilities and can penetrate at-
036 mospheric obstructions, their imagery is complex to inter-
037 pret, and publicly available datasets are scarce. Conversely,
038 widely available EO imagery is limited by lighting condi-
039 tions and atmospheric visibility. Domain translation seeks
040 to bridge this gap by converting one modality into another,
041 enabling cross-modal data synthesis and enhancing sensor
042 interoperability. By focusing on translation tasks such
043 as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the
044 MAVIC-T challenge aims to bolster existing data volume,
045 increase data diversity across regions, and serve as a plat-
046 form for advancing conditioned image synthesis techniques,
047 ultimately fostering the development of more adaptable and
048 robust models for multi-modal image analysis.

049 SAR images possess unique attributes that present chal-
050 lenges for both human observers and vision artificial intel-
051 ligence (AI) models to interpret, owing to their electromag-
052 netic (EM) characteristics [12–15, 23].

053 In this paper, we propose **EarthBridge**, a multi-method
054 framework for cross-modal aerial image translation. Our
055 primary approach is built on Diffusion Bridge Implicit
056 Models (DBIM) [48], which we generalize with non-
057 Markovian bridge processes to enable efficient determin-
058 istic sampling. In parallel, we explore Contrastive Un-
059 paired Translation (CUT) [27] as a powerful contrastive-
060 learning-based alternative. Unlike standard conditional dif-
061 fusion models that diffuse targets to pure noise and rely
062 on classifier-free guidance or simple concatenation for the
063 reverse path, EarthBridge implements a genuine stochastic
064 diffusion bridge. By directly constraining the forward
065 generative trajectory between the source modality x_T and
066 target modality x_0 , and parameterizing the UNet denoiser
067 to solve the bridge marginals under Karras-weighted scal-
068 ings [17], EarthBridge (DBIM) achieves high-fidelity trans-
069 lation across disparate modalities. We utilize pixel-level
070 modeling throughout both approaches to preserve the full
071 spectral and spatial resolution of the remote sensing data.

072 Our findings indicate that the deterministic DBIM sam-
073 pling significantly improves inference speed, achieving

high-quality results in as few as 5 sampling steps in
RGB→IR. Qualitatively, EarthBridge demonstrates a supe-
rior ability to reconstruct complex urban layouts and fine-
grained textures in challenging tasks such as SAR-to-RGB,
effectively bridging the domain gap. Quantitatively, our
solution achieves a composite task score of 0.269 for the
SAR→EO task and 0.38 overall, ultimately placing second
in the MAVIC-T challenge leaderboard.

Our main contributions are summarized as follows:

- We propose **EarthBridge**, a multi-method framework for paired cross-modal aerial image translation incorporating both Diffusion Bridge Implicit Models (DBIM) and Contrastive Unpaired Translation (CUT), providing a robust comparison between diffusion-based and contrastive-learning-based approaches.
- We develop a specialized training protocol using Karras-weighted bridge scalings and pixel-level modeling, optimized for the high dynamic range and resolution requirements of multimodal remote sensing data.
- We conduct comprehensive experiments on all four MAVIC-T tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), reporting competitive results and providing detailed quantitative and qualitative analysis.

2. Related Work

Translating across remote sensing (RS) modalities, such as SAR to optical (S2O), is critical for multisource data fusion. In this section, we provide a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing.

Image-to-Image and Cross-Modality Translation.

Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [16] and CycleGAN for unpaired translation [50]. Subsequent advances include contrastive learning for unpaired translation [27], zero-shot [28] and one-step translation with text-to-image models [29]. Translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [26], NIR imagery colorization [39], and deep colorization [23, 46].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [49] define a variance-preserving bridge between data distributions. Related formulations include

Brownian Bridge Diffusion Models (BBDM) [18] and its bidirectional extension BiBBDM [44], Image-to-Image Schrödinger Bridge (I^2SB) [19], Diffusion Schrödinger Bridge matching [37], Dual Diffusion Implicit Bridges [38], and Direct Diffusion Bridge with data consistency [4]. Recent extensions include Consistency Diffusion Bridge [8], Diffusion Bridge Implicit Models [48], Latent Bridge Matching for fast translation [2], deterministic translation via denoising Brownian bridge with dual approximators [43], and adaptive domain shift for cross-modality translation [42].

RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). Wang et al. [41] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [11] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation. Qin et al. [31] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [32] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Wang et al. [42] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. Chen et al. [3] introduced Any2Any, a unified framework for arbitrary modality translation in remote sensing, and highlighted its first use of Diffusion Transformers (DiT) in image-to-image translation. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y . In the context

of the MAVIC-T challenge, four translation tasks are defined (see Tab. 1): SAR→EO (1 → 1 channel, 256×256), SAR→RGB (1 → 3, 1024×1024), SAR→IR (1 → 1, 1024×1024), and RGB→IR (3 → 1, 1024×1024).

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS [47] (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual fidelity, FID [10] (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi} \arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \quad (1)$$

where the $\arctan$ normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of +1 added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

4. EarthBridge

We propose EarthBridge, built on Diffusion Bridge Implicit Models (DBIM) [48] for high-fidelity cross-modal translation. We also include Contrastive Unpaired Translation (CUT) [27] as a supplementary baseline for comparison.

4.1. Diffusion Bridge Implicit Models (DBIM)

Diffusion models have recently revolutionized generative modeling, yet they are fundamentally designed to transport complex distributions to a standard Gaussian prior. Adapting them to translate between two arbitrary distributions, such as heterogenous remote sensing modalities, requires conditioning the diffusion process on a fixed endpoint. This is achieved via Doob’s h -transform [7, 33].

Stochastic Bridges via h -transform. Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the conditioned SDE that almost surely reaches a prescribed target $x_0 = y$ at $t = 0$ starting from a source $x_T = x$ at $t = T$ is given by:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = y | x_t)dt, \quad (2)$$

where $p(x_0 = y | x_t)$ is the transition density of the original process. When the endpoints are fixed, the resulting conditioned process is called a *diffusion bridge* [5, 9, 20, 30, 35, 36], which forms the foundational mechanism for transporting one state to another. In the context of cross-modal translation, x represents the source modality (e.g.,

SAR) and y the target modality (e.g., optical). This formulation ensures that the generative trajectory is directly constrained by both endpoints, providing a more principled framework for paired translation than standard conditional diffusion.

Denoising Diffusion Bridge Models. Denoising Diffusion Bridge Models (DDBM) [49] provide a tractable realization of Eq. 2. Under a variance-preserving (VP) schedule, the forward bridge process constructs intermediate states z_t as:

$$z_t = a_t x + b_t y + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where a_t, b_t, σ_t are schedule-dependent coefficients. While DDBM excels in high-fidelity generation, its inference typically relies on simulating Eq. 2 via computationally expensive numerical solvers (e.g., Heun or stochastic samplers), often requiring hundreds of function evaluations (NFE) for optimal results. In this work, we adopt the more efficient *implicit* formulation to accelerate this process.

4.2. EarthBridge: Non-Markovian DBIM

We present our DBIM approach which generalizes the diffusion bridge framework with non-Markovian bridge processes defined on discretized timesteps. Our solution leverages the deterministic sampling property of DBIM to achieve high-fidelity cross-modal translation with significantly fewer sampling steps compared to the vanilla DDBM. Fig. 2 shows an overview of our EarthBridge pipeline.

Non-Markovian Diffusion Bridges. In contrast to the continuous-time Markovian bridge processes in DDBM, we consider a family of non-Markovian diffusion bridges defined over a sequence of discretized timesteps $0 = t_0 < t_1 < \dots < t_N = T$, controlled by a variance parameter $\rho \in \mathbb{R}^{N-1}$. The UNet denoiser is trained to predict the bridge marginals governed by the schedule-dependent coefficients (a_t, b_t, c_t), where the source image x is supplied to the network via channel concatenation. This concatenation is strictly an architectural mechanism to inject the mandatory endpoint parameter x into the SDE formulation, not an ad-hoc conditioning trick akin to standard image-to-image models. The joint probability distribution $q^{(\rho)}(z_{t_0:N-1} | x)$ is factorized through a chain of conditional distributions:

$$q^{(\rho)}(z_{t_0:N-1} | x) = q_0(z_{t_0}) \prod_{n=1}^{N-1} q^{(\rho)}(z_{t_n} | z_0, z_{t_{n+1}}, x), \quad (4)$$

where q_0 is the target modality distribution and each step $q^{(\rho)}$ is defined as:

$$\mathcal{N}(a_{t_n} x + b_{t_n} z_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \eta_{n+1}, \rho_n^2 I), \quad (5)$$

with $\eta_{n+1} = (z_{t_{n+1}} - a_{t_{n+1}} x - b_{t_{n+1}} z_0) / c_{t_{n+1}}$. Here, a_t, b_t, c_t are known coefficients determined by the VP log-SNR schedule (Sec. 4.1). Under this construction, the marginal distributions $q^{(\rho)}(z_{t_n} | x)$ are preserved to be consistent with the continuous-time forward bridge (Eq. (3)). The variance parameter ρ controls the level of stochasticity in the sampling process, ranging from a purely Markovian stochastic process to a fully deterministic implicit model.

Reverse Generative Process and Sampling. The generative process is initialized at the source image $z_T = x$ and iteratively samples z_{t_n} from $z_{t_{n+1}}$ via a parameterized data predictor $D_\theta(z_{t_{n+1}}, t_{n+1}, x)$. The core updating rule is given by:

$$z_{t_n} = a_{t_n} x + b_{t_n} \hat{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \hat{\epsilon} + \rho_n \epsilon, \quad (6)$$

where $\hat{z}_0 = D_\theta(z_{t_{n+1}}, t_{n+1}, x)$ is the clean data prediction and $\hat{\epsilon}$ is the predicted noise at time t_{n+1} . Setting $\rho_n = 0$ for all steps yields the deterministic *Diffusion Bridge Implicit Model (DBIM)*, which can be viewed as an Euler discretization of a probability flow ODE specifically tailored for diffusion bridges.

Booting Noise. A critical refinement in our sampling procedure is the introduction of *booting noise* at the initial step ($t_N = T$). Since $c_{t_N} = 0$, the denominator in $\hat{\epsilon}$ leads to singularity. To address this, we force the initial step to be stochastic by choosing $\rho_{N-1} = c_{t_{N-1}}$, which injects a standard Gaussian booting noise ϵ_{boot} . This booting noise serves as the latent variable, encoding the one-to-many mapping ambiguity in RS translation (e.g., several plausible infrared images for a single SAR input).

Channel Handling for Mismatched Modalities. For tasks where source and target modalities have different channel counts (e.g., RGB→IR or SAR→RGB), we operate the bridge in a shared model channels space (typically 3 for RGB-related tasks). Single-band modalities are expanded to model channels by channel repetition before entering the UNet, ensuring the concatenation $[z_t, x]$ remains well-defined across all tasks.

4.3. Contrastive Unpaired Translation (CUT) | Supplementary

As a supplementary baseline, we use CUT [27] for the SAR→IR task. Unlike diffusion bridges, CUT is a feed-forward GAN that performs translation in a single forward pass. It combines an adversarial loss with a patch-based contrastive loss to maintain structural correspondence between source and output. Architecture and hyperparameter details are in Sec. 6.1.

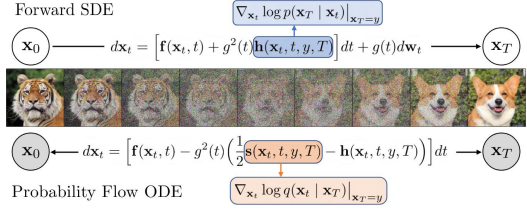
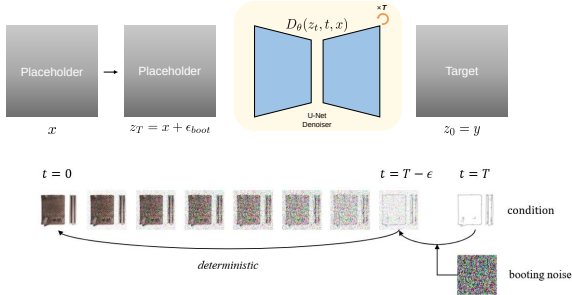
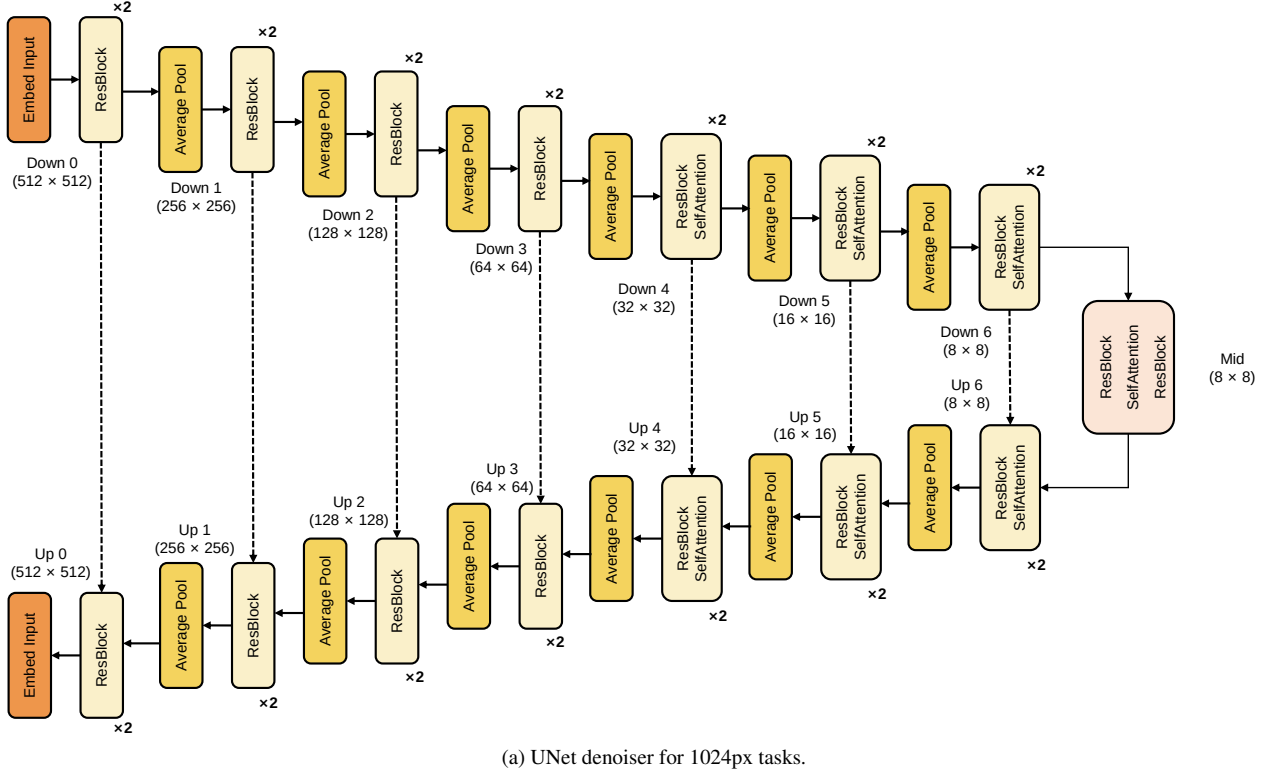


Figure 1: A schematic for Denoising Diffusion Bridge Models. DDBM uses a diffusion process guided by a drift adjustment (in blue) towards an endpoint $x_T = y$. They learn to reverse such a bridge process by matching the denoising bridge score (in orange), which allows one to reverse from x_T to x_0 for any $x_T = y \sim q_{\text{data}}(y)$. The forward SDE process shown on the top is unidirectional while the probability flow ODE shown at the bottom is deterministic and bidirectional. White nodes are stochastic while grey nodes are deterministic.

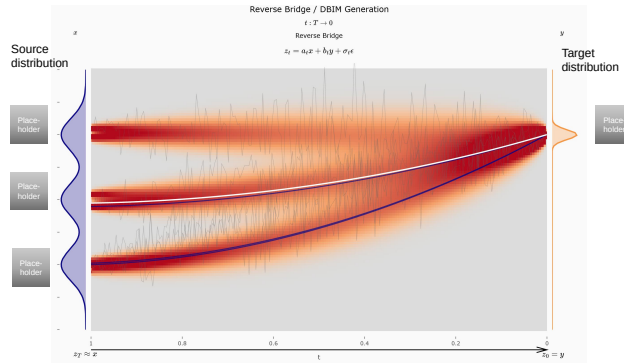


Figure 2. UNet architecture of our DBIM for 1024px tasks (SAR→RGB, RGB→IR). We train at 512×512 and infer at full resolution. Source x is channel-concatenated with noisy sample z_t and processed by this denoiser. Bridge scalings a_t, b_t, c_t construct z_t during training; deterministic DBIM sampling generates z_0 during inference.

Table 1. Task specifications and training data for the MAVIC-T challenge. Crop augmentation uses 1024×1024 sliding windows with stride 512 (50% overlap); SAR→optical pairs are georeference-aligned.

Task	Resolution	Channels	Samples (Orig.)	Samples (Aug.)
SAR→EO	256×256	1→1	89,411	—
SAR→RGB	1024×1024	1→3	10,576	75,703
SAR→IR	1024×1024	1→1	10,576	74,317
RGB→IR	1024×1024	3→1	2,272	7,388

PatchNCE Loss. CUT extracts multi-layer encoder features from both the source $\mathbf{x}$ and the generated image $\hat{\mathbf{y}} = G(\mathbf{x})$. A lightweight MLP samples spatial patches per layer, projects them to an embedding space, and applies L2 normalization. The PatchNCE (InfoNCE [40]) loss encourages that the query embedding from a patch in $\hat{\mathbf{y}}$ is closer to the key embedding from the corresponding patch in $\mathbf{x}$ than to embeddings from other patches. Formally, for query q and key k at the same spatial location, the loss uses $\ell_{\text{NCE}} = -\log \frac{\exp(q^\top k / \tau)}{\exp(q^\top k / \tau) + \sum_{k'} \exp(q^\top k' / \tau)}$, with temperature τ and negatives k' from other patches. This contrastive objective preserves structural layout across the translation.

Training Objective. The generator minimizes $\mathcal{L}_G = \lambda_{\text{GAN}} \mathcal{L}_{\text{GAN}} + \lambda_{\text{NCE}} \mathcal{L}_{\text{NCE}}$, where $\mathcal{L}_{\text{GAN}}$ is the LSGAN [24] objective (generator encourages the discriminator to classify $G(\mathbf{x})$ as real) and $\mathcal{L}_{\text{NCE}}$ is the per-layer PatchNCE loss averaged over the selected encoder layers.

5. Experiments

5.1. Dataset and Preprocessing

We use the MAVIC-T 2025 dataset from the challenge organizers, with imagery from UNICORN, USGS EROS HRO, and UMBRA across four regions: Bingham, Centerfield, Manhattan, and UC Davis. We do not use any external datasets beyond the official challenge data. Tab. 1 summarizes the four translation tasks. SAR→EO operates at 256×256 ; the other three tasks use 1024×1024 . For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. We also create crop-augmented training sets for the 1024×1024 tasks to increase data diversity.

5.2. Architecture and Training Objectives

Design Choice For DBIM-based tasks (SAR→EO, SAR→RGB, RGB→IR), we use a UNet [34] denoiser that predicts the clean target from the noisy sample conditioned on timestep and source image. For SAR→IR, we adopt CUT [27] as an alternative baseline, exploiting its patch-based contrastive learning for structural consistency. Per-task architectural and hyperparameter choices are summarized in Tab. 4.

Table 2. Main results on the MAVIC-T evaluation set. ↓ lower is better. Time is per-sample inference latency; SAR→EO is run at 256px with large batch size, and its latency is averaged.

Track	Method	FID↓	LPIPS↓	L1↓	Score↓	NFE	Time
SAR→EO	DBIM	0.22	0.50	0.08	0.27	500	0.42 s
SAR→RGB	DBIM	0.88	0.64	0.21	0.58	1000	160 s
SAR→IR	CUT	0.65	0.60	0.15	0.46	1	0.47 s
RGB→IR	DBIM	0.36	0.15	0.09	0.20	5	1.5 s

Table 3. Leaderboard comparison. Per-task scores (SAR→EO, SAR→RGB, SAR→IR, RGB→IR) from the challenge evaluation.

Rank	Participant	SAR→EO	SAR→RGB	SAR→IR	RGB→IR	Combined
1st	wangzhiyu918	0.11	0.50	<u>0.49</u>	0.20	0.32
2nd	EarthBridge (Ours)	<u>0.27</u>	<u>0.58</u>	0.46	0.20	<u>0.38</u>

Training Objective For DBIM, the primary training objective is a weighted mean squared error between the denoised prediction and the clean target:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(\mathbf{z}_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (7)$$

where $w(t)$ is the Karras bridge weighting. Diffusion time t is sampled from a Karras inverse- ρ distribution; inference uses deterministic DBIM sampling. Full technical details on architecture, schedule, and per-task configurations are in Sec. 6.1.

5.3. Quantitative Results

Tab. 2 presents the quantitative performance comparison between EarthBridge and the CUT baseline across the challenge tasks. Tab. 3 compares our second-place submission with the first-place result (wangzhiyu918); note that lower scores indicate better performance (the official leaderboard display was initially reversed). The composite task score provides a balanced evaluation of perceptual, distributional, and structural fidelity. EarthBridge demonstrates consistent performance across both low- and high-resolution tasks.

5.4. Qualitative Results

Fig. 3 shows qualitative examples across all four tasks. EarthBridge preserves fine-grained structural information from the source modality while accurately synthesizing target textures, demonstrating consistent performance across both low- and high-resolution settings.

6. Implementation Details

This section provides the comprehensive technical details of the EarthBridge architecture, training hyperparameters, and multi-modal refinement modules. We use the Prodigy optimizer [25] for all training. Training employs mixed precision in BF16; inference is also run in BF16.

6.1. Architecture, Sampling, and Scheduler Settings

DBIM. The denoiser uses an ADM-style UNet [6]. Source conditioning is via channel-wise concatenation

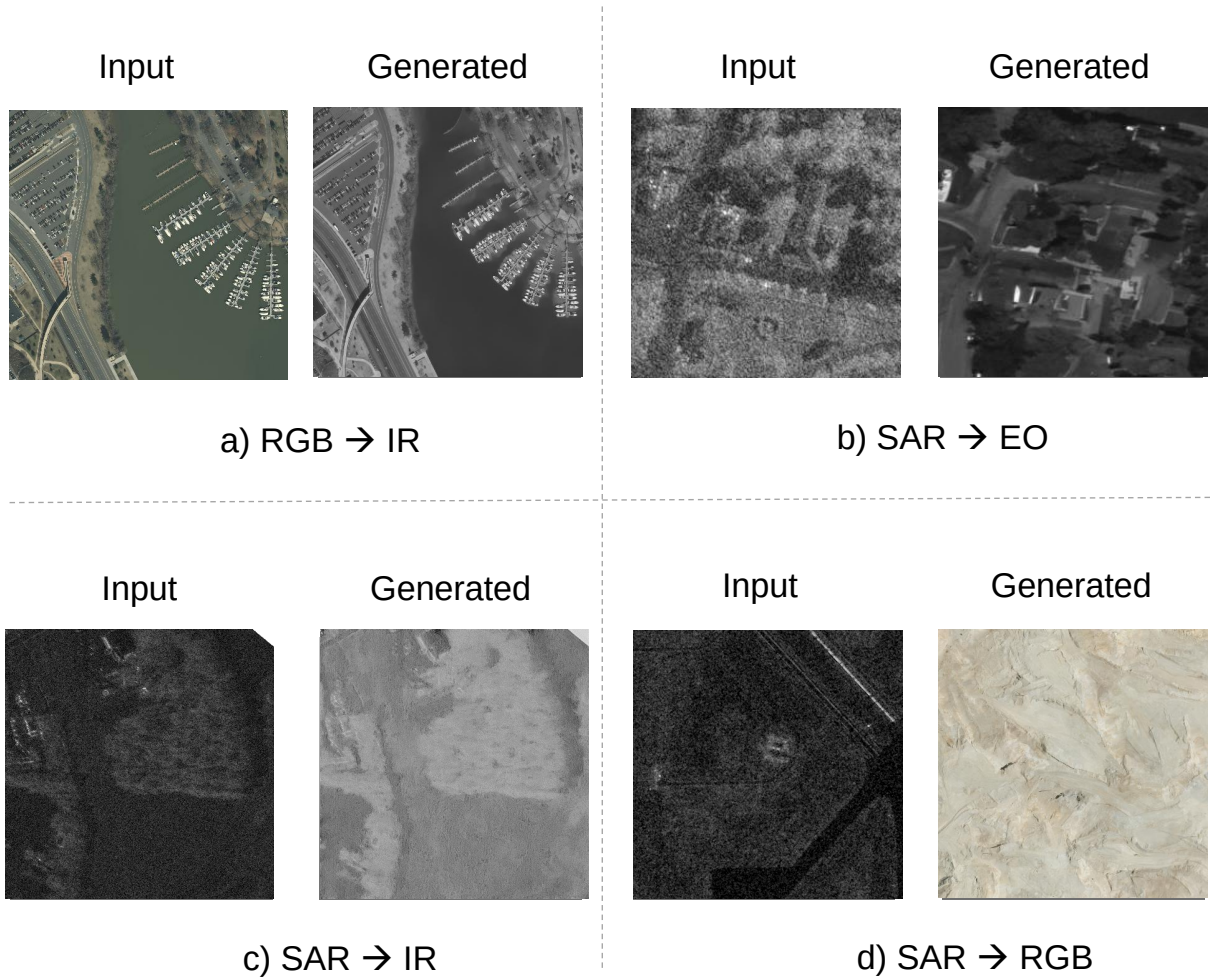


Figure 3. Qualitative results across all four MAVIC-T translation tasks. Each row corresponds to one task (SAR $\rightarrow$ EO, SAR $\rightarrow$ RGB, SAR $\rightarrow$ IR, RGB $\rightarrow$ IR). Columns show: source image, EarthBridge output, and ground-truth target. Our model preserves structural layout from the source while synthesizing faithful target-modality textures across both 256×256 and 1024×1024 resolutions.

[z_t, x]. For SAR $\rightarrow$ RGB, we use REPA [45] in our implementation, which relieves the loss spike issue in the early training steps. Sampling follows the deterministic DBIM formulation [48] with task-dependent inference steps (Tab. 2) and Gaussian booting noise ϵ_{boot} at $t = T$. Table 4 summarizes per-task architectural and training configurations, including the VP log-SNR schedule and Karras diffusion-time sampling.

CUT. The CUT formulation (PatchNCE loss, training objective) is described in Sec. 4.3. Our implementation follows the original architecture [27] with three core components: a ResNet-based generator, a PatchGAN [16] discriminator, and a PatchSampleMLP for patch-wise contrastive learning. Table 4 summarizes per-task architectural and training configurations.

7. Lessons and Insights

We summarize key lessons and insights from developing EarthBridge for the MAVIC-T translation track.

Diffusion Bridge vs. Standard Diffusion. The diffusion bridge directly builds a mapping from the source domain to the target domain, whereas standard diffusion starts from a noise prior (typically Gaussian noise) and requires a conditional generation strategy to accomplish this task. We hypothesize that the diffusion bridge is more efficient for training and achieves better results for cross-modal translation. However, standard diffusion can be more flexible for further conditional development, e.g., leveraging pre-trained models for a good prior. There is a trade-off between these approaches; we encourage readers to further explore both directions.

Table 4. Per-task architectural and training configurations. Top: DBIM settings for SAR→EO, SAR→RGB, and RGB→IR. Bottom: CUT settings for SAR→IR (ResNet-9block generator, 4-layer PatchGAN discriminator).

Configuration	SAR→EO	SAR→RGB	SAR→IR	RGB→IR
DBIM				
Parameters	~20M	~120M	—	~120M
VP schedule ($\beta_d, \beta_{\min}$)	2.0, 0.1	2.0, 0.1	—	2.0, 0.1
Karras ρ	7	7	—	7
Base channels	64	96	—	128
Channel multipliers	(1,2,3,4)	(1,1,2,2,4,4)	—	(1,1,2,2,4,4)
Attention resolutions	N/A	32, 16, 8	—	32, 16, 8
Training resolution	256 ²	512 ²	—	512 ²
Epochs	55	24	—	86
CUT				
Parameters (total)	—	—	~225M	—
Generator; Discriminator	—	—	~180M; ~45M	—
Generator (ResNet blocks)	—	—	9	—
Discriminator	—	—	4-layer PatchGAN	—
Filters (ngf, ndf)	—	—	256	—
GAN objective	—	—	LSGAN	—
$\tau, \lambda_{\text{GAN}}, \lambda_{\text{NCE}}$	—	—	0.1, 0.5, 1.0	—
Identity NCE	—	—	disabled	—
Training resolution	—	—	512 ²	—
Epochs	—	—	11	—

Resolution Scaling and Multi-Stage Curriculum. For 1024px modeling, we train on 512px only and directly infer on 1024px, which yields strong results while saving training compute budget. If time permits, we may further perform a quick fine-tuning on 1024px directly—we term this approach *multi-stage resolution curriculum learning*.

Task Difficulty and Input Modality. We found that SAR→EO and RGB→IR are substantially easier than the other two tasks. For RGB→IR, the three-channel input provides sufficient information for the translation. For SAR→EO, the relative ease may stem from both modalities originating from the same sensor, reducing the domain gap.

Inference Steps and Metric Trade-offs. Using the DBIM ODE solver, few-step inference yields visually appealing results. For RGB→IR, even a single-step generation is sufficient, and increasing steps tends to hallucinate details. In practice, however, many more inference steps

greatly improve FID and LPIPS in the competition metrics, even when L1 remains unchanged—highlighting a trade-off between perceptual quality, metric optimization, and inference efficiency.

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Supplementary Material